



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 – 2027

TITLE OF THE PROJECT	Intelligent Self-Healing Framework for Autonomous Software Fault Detection and Recovery	
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Abstract:

The Intelligent Self-Healing Framework for Autonomous Software Fault Detection and Recovery is designed to improve software reliability, availability, and performance by automatically detecting, diagnosing, and recovering from system faults with minimal human intervention. Traditional software systems often depend on manual monitoring and troubleshooting, which can increase downtime and delay fault resolution. The proposed framework continuously monitors application and system parameters such as CPU usage, memory utilization, response time, error rates, service status, and system logs to identify abnormal behavior.

The framework follows a detect, diagnose, recover, and verify approach. Intelligent fault detection techniques, including machine learning-based anomaly detection, are used to identify deviations from normal system behavior. After detecting a fault, the diagnostic module analyzes system information to determine its possible cause. The recovery module then automatically performs suitable corrective actions, such as restarting failed services, reallocating resources, rolling

back faulty changes, or restoring applications to a stable state. Finally, the framework verifies whether the recovery process has successfully restored normal operation.

The proposed system aims to reduce software downtime, improve recovery speed, minimize manual intervention, and enhance overall system resilience. It can be applied to cloud applications, web services, distributed systems, and other critical software environments requiring continuous availability and reliable autonomous operation.

Signature of the Guide

HOD